

FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

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The Patents Rules, 2003

COMPLETE SPECIFICATION (see section 10)

**CROWD CHAOS DETECTION SYSTEM FOR CROWD SAFETY MANAGEMENT
USING AUDIO-VISUAL SIGNAL**

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The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF INVENTION

25 The present invention relates to the fields of Artificial Intelligence, Signal Processing, Computer Vision, Crowd Analytics, Public Safety and Surveillance Systems, more specifically to the field of public crowd management for chaos detection by using artificial intelligence and audio and visual information collected in real time from the public area.

BACKGROUND OF THE INVENTION

30 Crowd behaviour is a fascinating field of sociology, analysing the behaviour of groups along the many aspects including psychology, social and societal, and study of the influence of the interactions on individuals and the society culturally, economically, and psychologically.

Several theories attempt to explain crowd behaviours, including convergence theory (crowds are composed of like-minded individuals), emergent norm theory (norms develop within the crowd), and behavioural contagion (imitation of others' actions). People generally exhibit herd mentality
35 in order to confirm to the perceived higher authority and increase their alignment towards them, in spite of their individual beliefs and behaviour. Deindividuation (loss of self-awareness and responsibility) and emergent norms (group-specific rules) are key factors influencing crowd behaviour.

Thus, a peaceful gathering can be incited to become violent, or panic can be induced with perceived
40 danger resulting in stampede or other destructive behaviour. Similarly, fraudulent activities like terrorism, pyramid schemes, cult formations rely on crowd analysis and crowd behaviour manipulation to increase their collectives. Study of crowd behaviour in entertainment business can also direct common people increasing the favour towards distinctive products, and people using different strategies.

45 Thus, to identify general crowd behaviour, presence of purposeful directive people to either induce buying behaviour or panic, and controlling crowds is essential during festivals, cultural programs, large gatherings at malls, grounds, processions, places of worship, working environments and so on.

Current systems either rely on video or audio individually, leading to limited reliability in chaotic
50 environments.

Audio analysis includes study of sound characteristics using the parameters like Mel-frequency cepstral coefficients (MFCCs), pitch, spectral flux, and harmonics-to-noise ratio (HNR), source identification, event detection, crowd volume and interplay.

Visual cues analysis includes study of crowd distribution, spatial analysis, movement, interactions, and event detection. There are different methods of combining the visual and audio cues to identify the crowd behaviour and event detection including machine learning techniques. Crowd analysis is useful for surveillance systems, urban planning, public safety, virtual reality, and simulation.

In the paper, “Ambient Sound Helps: Audiovisual Crowd Counting in Extreme Conditions” a dataset of visual images with corresponding audio is created using a linear feature-wise fusion module that carries out an affine transformation on visual and auditory features. This dataset can aid in crowd counting.

In the paper, “Audio–visual representation learning for anomaly events detection in crowds” relates to a two-branch network which includes a typical 3D CNN model to extract temporal appearance feature from video clips and an audio CNN model for encoding Log Mel-Spectrogram of audio signals. Finally, by fusing the above features, the more accurate prediction will be produced for crowd event detection.

In the paper, “Audio-visual multi-modality driven hybrid feature learning model for crowd analysis and classification” relates to a crowd classification model using a hybrid feature extraction model which was applied to extract deep spatial-temporal features by using Gray-Level Co-occurrence Metrics (GLCM) and AlexNet transferrable learning model. Horizontal concatenation was done to fuse the different feature sets extracted from these models. Similarly, for acoustic feature extraction, the audio samples (from the input video) were processed for static (fixed size) sampling, pre-emphasis, block framing and Hann windowing, followed by acoustic feature extraction like GTCC, GTCC-Delta, GTCC-Delta-Delta, MFCC, Spectral Entropy, Spectral Flux, Spectral Slope and Harmonics to Noise Ratio (HNR). Finally, the extracted audio-visual features were fused to yield a composite multi-modal feature set, which is processed for classification using the random forest ensemble classifier.

The patent “Audio Event Detection and Localization System” (US9812150B2) relates to a system for detecting and localizing audio events using distributed microphone arrays, and is focused on gunshot detection and not optimized for crowd dynamics.

The patent “Crowd Behavior Analysis and Monitoring System Using Acoustic Sensors” (US10134271B1) relates to a system which uses distributed acoustic sensors to detect and classify crowd behaviors and potential security threats, but it is limited to acoustic analysis only, no visual integration or advanced harmonics fingerprinting.

The patent “multi-dimensional crowd characteristic measuring method” (CN111460245A) relates to a system which utilises multidimensional information for predicting or judging potential behaviour tendency of target personnel and receivable behaviour service or communication mode. This model is more geared towards identification of specific persons in crowd and analysing their behaviour.

Thus, several models exist on obtaining audio/visual cues from the ground regarding crowds, and analysing, categorising and identifying issues pertaining to the crowds/ individuals in crowds

Our present invention proposes a novel integrated architecture that combines harmonics-based sound fingerprinting, emotional facial detection, and crowd density estimation from both audio and video datasets. The system introduces a mathematical risk scoring framework that fuses all multi-modal signals (emotion, crowd density, and audio) into a single, interpretable " Risk Score" for real-time safety interventions.

OBJECTIVE OF THE INVENTION

The primary objective of this invention is to detect crowd density and predict potential chaotic scenario events through emotional behaviour and audio cues using Multi-modal(audio, video) based chaotic crowd analysis and Harmonics-Based Sound Fingerprinting (HBSF) for crowd noise differentiation and Knowledge distillation and YOLOv8n-face technique for detecting faces for emotional behavioural insights.

Another objective of our present invention is to detect crowd density using both audio and video in real-time including Crowd Acoustic Density Analysis (CADA) for audio-only crowd estimation.

Another objective of our present invention is to enhance public safety by deploying a multimodal surveillance mechanism.

Another objective of our present invention is to build a system robust to occlusions, low visibility, and noisy environments.

110 SUMMARY OF THE INVENTION

The following summary is provided to facilitate a clear understanding of the new features in the disclosed embodiment and it is not intended to be a full, detailed description. A detailed description of all the aspects of the disclosed invention can be understood by reviewing the full specification, the drawing and the claims and the abstract, as a whole.

115 The present invention relates to a crowd detection and management system with an integrated architecture that combines harmonics-based sound fingerprinting, emotional facial detection, and crowd density estimation from both audio and video datasets. The system introduces a mathematical risk scoring framework that fuses all multi-modal signals (emotion, crowd density, and audio) into a single, interpretable " Risk Score" for real-time safety interventions.

120 The present invention relates to a system wherein the Visual Detection and Tracking module consists of YOLOv8x.pt for person detection and YOLOv8n-face.pt for emotional behaviour in crowd chaos monitoring. The Acoustic Direction and Intensity Analysis module of our present system includes directional microphones to capture ambient sound, analyzed using Fast Fourier Transform(FFT) to generate polar plots identifying predominant crowd sound direction. The third
125 module consists of calculating chaotic risk score using individual weights from Emotion(E_Score), Density(D_Score), Audio chaos(A_Score) Visual Detection and Tracking. Together, the modules work together to perform a comprehensive analysis of the crowd gathering and present the detailed report to the monitoring agents.

BRIEF DESCRIPTION OF THE DRAWINGS

130 The manner in which the present invention is formulated is given a more particular description below, briefly summarized above, may be had by reference to the components, some of which is illustrated in the appended drawing It is to be noted; however, that the appended drawing illustrates

only typical embodiments of this invention and are therefore should not be considered limiting of its scope, for the system may admit to other equally effective embodiments.

135 Throughout the drawings, the same drawing reference numerals will be understood to refer to the same elements and features.

The features and advantages of the present invention will become more apparent from the following detailed description a long with the accompanying figures, which forms a part of this application and in which:

140 **Fig 1:** Diagram showing the system architecture, in accordance with our present invention;

Fig 2: Diagram showing the experimental results, wherein: Fig 2.1 shows the Sample captured result conveying chaotic risk case namely “caution”, Fig 2.2 shows the Sample captured result conveying chaotic risk case namely “warning” and Fig 2.3 shows the Sample captured result conveying chaotic risk case namely “safe”. in accordance with our present invention;

145 **REFERENCE NUMERALS**

101 – Audio Model

102 – Crowd Video Dataset

103 – Preprocessing

104 – Conversion of Audio file from .mp4 to .wav

150 105 – Acoustic Quality Enhancement

106 – Extracting Audio Fingerprint Characteristics using Harmonic Fingerprint Extraction

107 – Detecting Audio direction and crowd density determination using Crowd Acoustic Density Analysis (CADA)

108 – Audio Direction Visualization using 3D Beamforming Technique

155 109 – o/p: audio direction

110 – o/p: Acoustic Score (A_Score) and Direction

	111 – Crowd Movement Flow Direction
	112 – Teacher Model
	113 – Crowd Video Dataset
160	114 – Preprocessing
	115 – Feature Extraction using Convolutional Neural Network (CNN)
	116 – Human Detection using YoloV8x
	117 – o/p: Person Detection with Bounding Boxes
	118 – Cascaded Knowledge Distillation
165	119 – Person Detection (Bounding Boxes received from Teacher Model)
	120 – Student Model
	121 – Crowd Video Dataset
	122 – Face Detection Model
	123 – Pre-processing
170	124 – Detecting Faces Using YoloV8n-Face
	125 – Cropping Faces from Images
	126 – o/p: Individual Faces
	127 – Emotion Detection Model
	128 – Convolutional Neural Network (CNN)
175	129 – Detecting Emotions Using YoloV8n-Face
	130 – o/p: Bounding Boxes with Faces, Person and Emotion
	131 – o/p: Crowd Emotion Score (E_Score) and Emotion Distribution Values for Neutral, Fear, Anger, Surprise, Sad, Happy, Disgust
	132 – o/p: Crowd Density Score (D_Score)

180 133 – Chaotic Risk Score = $0.4E_Score + 0.4D_Score + 0.2*A_Score$

134 – Chaotic Risk Score ≤ 0.4

135 – Safe Notification

136 – Stop

137 – Chaotic Risk Score < 5

185 138 – Emotion Notification

139 – Stop

140 – Chaotic Risk Score < 7

141 – Warning Notification

142 – Stop

190 143 – Critical Notification

144 – Send notification to water sprinklers for crowd to recover from suffocations

145 – Stop

DETAILED DESCRIPTION OF THE INVENTION

195 The principles of operation, design configurations and evaluation values in these non-limiting examples can be varied and are merely cited to illustrate at least one embodiment of the invention, without limiting the scope thereof.

The embodiments disclosed herein can be expressed in different forms and should not be considered as limited to the listed embodiments in the disclosed invention. The various embodiments outlined in the subsequent sections are constructed such that it provides a complete and a thorough understanding of the disclosed invention, by clearly describing the scope of the invention, for those skilled in the art.

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Throughout this specification various indications have been given as to preferred and alternative embodiments of the invention. It should be understood that it is the appended claims, including all equivalents, which are intended to define the spirit and scope of this invention.

205 The present invention relates to a system with Harmonics-Based Sound Fingerprinting (HBSF) for crowd noise differentiation, Crowd Acoustic Density Analysis (CADA) for audio-only crowd estimation, multi-modal(audio, video) based chaotic crowd risk scoring formula, Knowledge distillation and YOLOv8n-face technique for detecting faces for emotional behavioural insight and a automated Emergency sprinkler response integration system for crowd management.

210 The present invention details a Crowd Risk Assessment System that utilizes integrated audio, visual, and machine learning models to calculate a Chaotic Risk Score (CR_Score) and trigger graded notifications. The system is comprised of an Audio Model (101), a Teacher Model (112), and a Student Model (120), all feeding into a unified risk calculation and notification logic.

The Audio Model (101) processes acoustic data to determine an Acoustic Score (A_Score) (109) and Direction (110). The process begins with a Crowd Video Dataset (102) as the source material. 215 Preprocessing (103) converts the audio file from its original format to a .wav (104) file, followed by Acoustic Quality Enhancement (105). Audio characteristics are then extracted using Harmonic Fingerprint Extraction (106). Crowd Acoustic Density Analysis (CADA) (107) detects audio direction and visualizes crowd density, and an Audio Direction Visualization (108) using a 3D 220 Beamforming Technique is applied. The outputs are the Acoustic Score (A_Score) (109) and the audio Direction (110).

The Teacher Model (112) processes video data to derive a Crowd Density Score (D_Score) (132) and Crowd Movement (111). The model uses a Crowd Video Dataset (113) and applies Preprocessing (114) to the video stream. Feature Extraction (115) is performed using a 225 Convolutional Neural Network (CNN). Human Detection (116) employs YOLOv8 to identify individuals, generating an output of Person Detection and Bounding Boxes (117). These outputs contribute to the determination of Crowd Movement (111) and the final Crowd Density Score (D_Score) (132).

The Student Model (120) focuses on detailed, individual-level analysis, specifically Face 230 Detection (122) and Emotion Detection (127), using the Crowd Video Dataset (121). Prior to processing, the Student Model (120) receives Person Detection and Bounding Boxes (118) from the Teacher Model (112) through Cascaded Knowledge Distillation (119). The Face Detection Model (122) first performs Pre-processing (123), followed by Detecting Faces (124) using

YOLOv8n-Face, and finally Cropping Faces from Images (125) to isolate Individual Faces (126).
 235 The Emotion Detection Model (127) takes the Individual Faces (126) as input and utilizes a Convolutional Neural Network (CNN) (128) for classification. Detecting Emotions (129) using YOLOv8n-Face results in the output of Bounding Boxes with Faces, Poses, and Emotion (130). The Student Model's (120) final output is the Crowd Emotion Score (E_Score) and Emotion Distribution Values (131) for various states (Neutral, Fear, Anger, Surprise, Sad, Happy, Disgust).
 240 The system calculates the Chaotic Risk Score (CR_Score) (133) using the following weighted formula: $CR_Score = 0.6 \times D_Score + 0.4 \times E_Score + 0.2 \times A_Score$. This score is then passed through a sequential decision-making process to trigger appropriate notifications. The process evaluates the score against multiple thresholds. If the Chaotic Risk Score ≥ 35 (134) is No, a Safe Notification (135) is triggered, resulting in a Safe (136) status. If the condition Chaotic Risk Score ≥ 35 (134)
 245 is Yes, the system checks the next threshold: Chaotic Risk Score ≥ 55 (137). If this condition is No, a Moderate Notification (138) is triggered, resulting in a Moderate (139) status. If the condition Chaotic Risk Score ≥ 55 (137) is Yes, the system checks the next threshold: Chaotic Risk Score ≥ 70 (140). If this is No, a Warning Notification (141) is triggered, resulting in a Warning (142) status. If the final condition Chaotic Risk Score ≥ 70 (140) is Yes, a Critical Notification (143) is
 250 triggered. This critical state activates an emergency response to Send notification to water sprinklers for crowd to recover from riot/suffocation (144), leading to the Critical (145) status.

The system of our present invention operates in three primary stages:

1. Visual Detection and Tracking: YOLOv8x.pt for person detection and YOLOv8n-face.pt for emotional behaviour in crowd chaos monitoring.

255 **2. Acoustic Direction and Intensity Analysis:** Directional microphones capture ambient sound, analyzed using Fast Fourier Transform(FFT) to generate polar plots identifying predominant crowd sound direction (101).

3. Calculating chaotic risk score using individual weights from Emotion(E_Score),

Density(D_Score), Audio chaos(A_Score) using the following formula:

260 ***RiskScore = 0.4 \times E_Score + 0.4 \times D_Score + 0.2 \times A_Score***

System Components of our present invention:

The components of one embodiment of our present invention are given by:

1. Video-Based Crowd Chaos Detection:

265 Camera feed processed using YOLOv8x.pt model for person detection and human identification

YOLOv8n-face.pt model used specifically for face detection and emotion recognition

Individual faces cropped from images for emotion analysis

Continuous monitoring of person counts and emotional behaviour patterns

2. Audio-Based Crowd Chaos Detection:

270 Crowd audio signals are pre-processed using Harmonic Fingerprint Extraction technique

Crowd Acoustic Density Analysis (CADA) estimates crowd density from audio patterns

Acoustic Quality Enhancement and 3D Beamforming for detecting sound direction analysis

Outputs Acoustic Score (A_Score) and Direction for spatial crowd movement detection

275 3. Student Teacher Knowledge Distillation for Face Emotion Detection:

Teacher Model: Processes crowd video dataset with preprocessing and feature extraction using Convolutional Neural Networks (CNN)

Student Model: Utilizes crowd video dataset with faces for individual person detection and emotion analysis

280 Knowledge Distillation with weight training to detect faces of individual persons

Convolutional Neural Network (CNN) for emotion detection using YOLOv8n-Face model

4. Risk Assessment and Decision Making:

Chaotic Risk Score Calculation: Chaotic Risk Score = $0.4 \times E_Score + 0.4 \times D_Score + 0.2 \times A_Score$

285 Case 1 (127) (Safe Notification): Risk Score < 35

Case 2 (130) (Caution Notification): Risk Score < 55

Case 3 (133) (Warning Notification): Risk Score < 75

Case 4 (136) (Critical Notification): Risk Score > 75

290 Automated Emergency Response: Critical situations trigger water sprinkler systems for crowd safety-management

In one embodiment, the system is directed towards processing the audio component of crowd videos to extract density, direction, and intensity information. The method begins with audio pre-processing in which the sound track embedded in an .mp4 video is isolated and converted into an uncompressed .wav format, thereby preserving time–frequency fidelity required for further analysis. The converted audio signal undergoes acoustic quality enhancement through noise suppression, reverberation reduction, and adaptive filtering techniques that reduce environmental interference while improving clarity of human vocalizations and crowd-generated sounds. From this enhanced signal, harmonic fingerprint extraction is performed by deriving chroma features that quantify the energy distribution across pitch classes, and by constructing spectrograms using short-time Fourier transforms to visualize temporal frequency content. High-energy peaks within the spectrograms are detected and mapped to form unique harmonic patterns that serve as identifying acoustic fingerprints of collective crowd activity. Subsequently, a Crowd Acoustic Density Analysis (CADA) engine evaluates multiple signal descriptors including spectral centroid, spectral roll-off, root mean square (RMS) energy, zero-crossing rate (ZCR), Mel-frequency cepstral coefficients (MFCCs), tempo variation, onset rate, and spectral contrast. These features are processed through a scoring model to generate a normalized CADA Score ranging from 0 to 100, which corresponds to a Crowd Density Level classified as Very Low, Low, Medium, High, or Very High. Concurrently, audio direction estimation is conducted by applying spatial audio analysis methods such as interaural phase difference and beamforming to determine the dominant sound source location relative to the recording device. The directional output is provided in

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compass-based terms (N, NE, E, SE, S, SW, W, NW), along with a confidence score reflecting accuracy of detection. The outputs of this embodiment therefore comprise an Acoustic Score (A_Score) representing the magnitude of crowd acoustic activity, a directional and angular orientation of dominant sound sources, and a qualitative Crowd Density Level derived from CADA, enabling real-time assessment of crowd dynamics solely through audio analysis.

In another embodiment, the system implements a student - teacher knowledge distillation framework for image-based detection of individual faces and recognition of emotional states within a crowd. The teacher model, trained using convolutional neural networks (CNNs) on large annotated crowd video datasets, processes incoming frames to identify human figures and generates bounding boxes that encompass each detected person. These bounding boxes are used as supervisory guidance for the student model, which specializes in high-resolution face detection and emotion analysis. The student model employs a YOLOv8n-Face detector restricted to searching within the teacher-provided bounding boxes, ensuring that for every person annotated by the teacher, the student attempts to localize a corresponding facial region. Through cascaded knowledge distillation, the teacher's coarse-level person detection is translated into fine-level face localization by the student, thereby improving detection reliability in dense and partially occluded crowd environments. Once a face is detected, the student model further classifies the emotional state of the individual using feature extraction from localized facial regions and applying classifiers trained on emotion-annotated datasets. This approach ensures that the knowledge of the teacher model is systematically transferred to the student model, achieving both efficiency and robustness in detecting multiple faces and identifying their associated emotions in real time. The combined output of this embodiment includes the bounding box of each individual person, the refined face detection within that region, and the classified emotion, thus enabling integration of person-wise affective states with audio-derived metrics for comprehensive multi-modal crowd behaviour monitoring.

The metric table Table 1 shows the scores and how their values are assigned. The table Table2 shows the weightage assigned to different emotions. The table 3 shows the risk score classification along with the system action.

Table 1

Metric	What it measures	Core formula	Key novelty / significance
E_Score	Emotional stress level from facial expressions	$E_Score = \frac{\sum_{i=1}^{n_{faces}} w_i \cdot e_i}{n} \times 10$ where: e_i : is the normalized percentage of person showing emotion i w_i : is the weight in the table above - n: is the number of faces detected	First real-time, <i>weighted</i> emotion quantification for crowd-risk prediction
D_Score	Person density vs. venue safety limit	$D_Score = \min \left[100, \left(\frac{person\ count}{\theta} \right) \times 100 \right]$	Adaptable density threshold enables venue-specific tuning
A_Score	Audio chaos from crowd-noise patterns	$A_Score = \min(100, 0.4I + 0.3F + 0.3S)$ Where : I- Audio Intensity F- Movement fluctuation S- Ambience only sound	Introduces CADA to fold 3-mic audio features into a single chaos metric

Table 2

Emotion	Weight (W_i)
Fear	10
Anger	9
Surprise	7
Disgust	6
Sad	4
Not Neutral	2
Happy	1

Table 3

Risk range	Classification	System action
<35	Safe	Routine monitoring
<55	Caution	Heightened watch
<75	Warning	Proactive Chaos Avoidance Measures
>75	Critical	Sprinkler activation

345 **Experimental results of one embodiment of our present invention:**

1. Sample captured result conveying chaotic risk case namely “caution”

The experimental results of this sample are given below and the fig2.1

Persons detected = 15 - Faces detected = 5

Emotion distribution → Neutral 40%, Fear 25%, Anger 15%

350 E_Score = 61.5 D_Score = 50.0 A_Score = 0.0

Final Risk = 44.6 / 100 → Caution (monitor closely)

System status = All modules passed; no emergency action triggered

2. Sample captured result conveying chaotic risk case namely “warning”

The experimental results of this sample are given below and the fig2.2

355 Persons detected = 13 - Faces detected = 13

Emotion distribution → Neutral 40%, Fear 25%, Anger 15%

E_Score = 61.5 D_Score = 50.0 A_Score = 68.2

Final Risk = 58.2 / 100 → Warning (immediate crowd-control intervention issued)

Audio direction = NW (325.1 °) with 121% confidence

360 Emergency routine activated as per risk-threshold logic

3. *Sample captured result conveying chaotic risk case namely “safe”*

The experimental results of this sample are given below and the fig 2.3

Persons detected = 19 - Faces detected = 0 (no emotion data)

E-score = 0.0 D-score = 70.0 A-score = 0.0

365 Final Risk = 28.0 / 100 → Safe (continue standard surveillance)

System confirms low emotional stress despite moderate density

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380 **CLAIMS:**

I/We claim:

1. A system for real-time chaotic crowd detection and management, comprising:

- 385 a. An Audio Model (101) configured to process a Crowd Video Dataset (102) audio stream, said processing including: i. Preprocessing (103) for conversion to a .wav (104) file; ii. Acoustic Quality Enhancement (105); iii. Harmonic Fingerprint Extraction (106) for extracting audio fingerprint characteristics; and iv. Crowd Acoustic Density Analysis (CADA) (107) and 3D Beamforming Technique (108) for generating an Acoustic Score (A_Score) (109) and Direction (110);
- 390 b. A Teacher Model (112) configured to process a Crowd Video Dataset (113) video stream, said processing including: i. Preprocessing (114) and Feature Extraction (115) using a Convolutional Neural Network (CNN); and ii. Human Detection (116) using YOLOv8 to generate Person Detection and Bounding Boxes (117), a Crowd Movement (111) output, and a Crowd Density Score (D_Score) (132);
- 395 c. A Student Model (120) configured to perform individual-level analysis, said model receiving Person Detection and Bounding Boxes (118) from the Teacher Model (112) via Cascaded Knowledge Distillation (119), and comprising: i. A Face Detection Model (122) using YOLOv8n-Face (124) for detecting and Cropping Faces (125) to isolate Individual Faces (126); and ii. An Emotion Detection Model (127) using a Convolutional Neural Network (CNN) (128) and YOLOv8n-Face (129) to output Bounding Boxes with Faces, Poses, and Emotion (130) and a Crowd Emotion Score (E_Score) (131);
- 400 d. A Risk Scoring Module configured to compute a Chaotic Risk Score (CR_Score) (133) according to the formula: $CR_Score = 0.6 \times D_Score + 0.4 \times E_Score + 0.2 \times A_Score$;
- 405 e. A Decision and Emergency Response Module configured to classify the Chaotic Risk Score (133) based on defined thresholds: i. Triggering a Safe Notification (135) if the score is less than a first threshold (e.g., $CR_Score < 35$) (134); ii. Triggering a Moderate Notification (138) if the score is greater than or equal to the first threshold (134) and less than a second threshold (e.g., $CR_Score < 55$) (137); iii. Triggering a Warning Notification (141) if the score is greater than or equal to the second threshold (137) and less than a third threshold (e.g., $CR_Score < 70$) (140); and iv. Triggering a Critical Notification (143) and

- 410 an automated command to Send notification to water sprinklers (144) if the score is greater than or equal to the third threshold (140).
2. The system of Claim 1, wherein the Harmonic Fingerprint Extraction (106) module generates harmonic-based sound fingerprints (HBSF) for differentiating chaotic crowd noise from
415 ambient sounds.
3. The system of Claim 1, wherein the Crowd Acoustic Density Analysis (CADA) (107) module is further configured to estimate crowd density solely from audio signals when visual inputs are unavailable due to occlusions or low visibility.
- 420 4. The system of Claim 1, wherein the Risk Scoring Module applies the predefined weightage of $0.6 \times D_Score$, $0.4 \times E_Score$, and $0.2 \times A_Score$ to calculate the Chaotic Risk Score (CR_Score) (133).
- 425 5. The system of Claim 1, wherein the Decision and Emergency Response Module classifies the Chaotic Risk Score (133) into categories corresponding to the thresholds: Safe (136), Moderate (139), Warning (142), and Critical (145), each category being mapped to a corresponding automated system action.
- 430 6. The system of Claim 1, wherein the Emotion Detection Model (127) using YOLOv8n-Face (129) is configured to classify facial emotions into categories including Neutral, Fear, Anger, Surprise, Sad, Happy, and Disgust, and to compute the Crowd Emotion Score (E_Score) (131) based on the emotion distribution.
- 435 7. The system of Claim 1, wherein the Cascaded Knowledge Distillation (119) transfers detection and bounding box information (118) from the Teacher Model (112) to the Student Model (120) to enhance the robustness of face and emotion recognition.
- 440 8. The system of Claim 1, wherein the Audio Model (101) performs Audio Direction Visualization (108) using the 3D Beamforming Technique to enhance the spatial resolution of crowd sound detection.

9. The system of Claim 1, wherein the emergency response action (144) automatically triggers a water sprinkler system when the Chaotic Risk Score (133) is classified as Critical (145).

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10. The system of Claim 1, wherein the system is configured to generate an interpretable risk score dashboard for display to monitoring agents, integrating the Acoustic Score (A_Score) (109), Crowd Density Score (D_Score) (132), and Crowd Emotion Score (E_Score) (131).

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A CROWD CHAOS DETECTION SYSTEM FOR CROWD SAFETY MANAGEMENT USING AUDIO-VISUAL SIGNALS

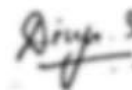
ABSTRACT

The present invention is a novel Crowd Risk Assessment System that integrates multimodal data for real-time chaotic crowd detection and management. The system utilizes three core models. The Audio Model (101) uses Harmonic Fingerprint Extraction (106) and 3D Beamforming (108) to generate an Acoustic Score (A_Score) (109). The Teacher Model (112) employs YOLOv8 (116) for person detection, determining the Crowd Density Score (D_Score) (132). The Student Model (120), linked via Cascaded Knowledge Distillation (119), uses YOLOv8n-Face (124, 129) for detailed emotion analysis, yielding the Crowd Emotion Score (E_Score) (131). A central Risk Scoring Module calculates the Chaotic Risk Score (CR_Score) (133) as a weighted sum: $0.6 \times D_Score + 0.4 \times E_Score + 0.2 \times A_Score$. The system then employs a Decision Module to classify the risk into levels (e.g., Safe (136) to Critical (145)), triggering appropriate notifications and, for critical risks, an automated response, such as a water sprinkler system (144).

Fig. 1

Dated this 18th day of August 2025

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